

Vivarium pitch

AniPixel Productions



Team

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Overview

The main aspects of our game “Vivarium”:

- The game offers the user the ability to choose between different creatures and environments. *Alien*, *Vampire* and *Bird* are the creature options that come with the first version of the game. The user can then choose the habitat for the selected creature from the following environments: *Forest*, *Desert* and *Antarctica*.
- The game has a vitals system, where *health* and *hunger* are shared amongst all of the creatures. Additionally, every creature will have a 3rd unique vital. Vampire has *photosensitivity*, meaning that too much sunlight exposure could be fatal for it. Secondly, the bird has the *flight* vital, which indicates the stamina left for flying. Finally, the alien has the *shape-shift* vital which shows the amount of times the creature could transform their shape into another one. Changing shapes rapidly will cause a cooldown where the user will have to wait before they can change the alien's shape again.
- The main type of stakeholders are children from the age of 4 and above, because they are the ones that will interact with the system. Another type of stakeholders are the parents of the children which can give this game to kids instead of buying them a real life pet. The other stakeholder is our team *AniPixel Productions* that is developing the game together with our supervisor that oversees the team's work.
- Since our target audience is children, there will be a basic graphical user interface for easy and intuitive interaction with the game.
- It is a video game in which you can take care of your creature. You have to feed it, play with it to keep it happy, make it sleep. When the creature is not fed or happy, it dies.
- Inspiration for this project was taken from the popular game Pou, <http://www.pou.me/>.
- One of the libraries that we will use is JavaFX, which is crucial to the graphical part of the project.

Functional features

ID	Short name	Description	Champion
F1	Pet care	The player should be able to take care of the pet by feeding it. The way they can earn money for food is by playing a mini game. If all of the vitals percentages are below a certain threshold, the health bar will start depleting. If the player doesn't prevent the health bar from reaching 0, the pet creature will immediately die. This will be conveyed to the user by displaying a "Game Over" message to them. One way the user can revive the pet is by watching an ad from our sponsors.	Kosta
F2	Status indicators	The GUI shall display status bars indicating the level of that attribute. We are going to have: - Health bar (All) - Hunger bar (All) - Photosensitivity bar(Vampire) - Shape-shift bar(Alien) - Flight(Bird)	Ibrahim
F3	Shop system	The earned money from playing the mini-game can be spent on food in the shop to sustain the health of the pet creature	Darian
F4	Minigame	The player can play a minigame to earn coins and increase the health percentage. The more riddles the player can guess, the more coins they can earn.	Diar
F5	Environment system	The user will have the ability to choose from various environments for their pet creature. All of them will have different sunlight intensity which will affect the vampire's photosensitivity.	Darian

Quality requirements

ID	Short name	Quality attribute	Description
QR1	Buttons	Responsiveness	When a button is clicked with the mouse, the result should appear on the screen within 0.3 seconds.
QR2	Extensibility	Maintainability	The system shall be structured in such a way that new types of characters or new minigames can be added to the system without affecting the other parts of the system.
QR3	Interface	Usability	The interface is easy to understand and use. The buttons for all of the functions are relatively large and their functions are intuitive from the appearance. Moreover, the buttons are placed in convenient places on the screen for easy reachability.
QR4	Game mechanics	Reliability	Graphics, user input, and audio should be working as expected at all times without crashing the game.

Time log

Team Number:	22		
Member	Activity	Week number	Hours
-Darian Janevski -Diar Kamberi -Kosta Bakalinov -Ibrahim A. I. Badr	Kick start meeting	1	3
Diar Kamberi	Overview and libraries	1	2
Ibrahim A. I. Badr	Quality Requirements	1	2
Darian Janevski	Define functional features	1	2
Kosta Bakalinov	Overview and research	1	2
-Darian Janevski -Diar Kamberi -Kosta Bakalinov -Ibrahim A. I. Badr	Meeting with TA and then making changes to the assignment based on the TA's feedback	2	4
-Darian Janevski -Diar Kamberi -Kosta Bakalinov -Ibrahim A. I. Badr	Discussing and finishing the team contract and time logs	2	2
		Total	17

Signed contract

https://drive.google.com/file/d/1gn03_vPlkLt9uobyR4xfflRE5OdcI1yo/view?usp=share_link